

Triangularization and Duality of the Normalized IQC System

Let P be the primal input–output operator and let

$$\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix}, \quad J_{z,w} = \begin{bmatrix} I_z & 0 \\ 0 & -I_w \end{bmatrix}, \quad J_{w,z} = \begin{bmatrix} I_w & 0 \\ 0 & -I_z \end{bmatrix}.$$

Assume that Ψ is boundedly and causally invertible. To use the triangularization of Yuan and Wu, assume additionally that Ψ_{22} has a bounded causal inverse. Finally, assume that

$$F_0 := \Psi_{21}P + \Psi_{22}$$

has a bounded causal inverse. This is the input–output counterpart of the current assumption in Lemma 1, $(\mathcal{D}_{\Theta}^{wz}\mathcal{D} + \mathcal{D}_{\Theta}^w)^{-1} \in \Pi_4$. Throughout, $\top$ denotes the causal dual of an input–output operator; $*$ is retained for realization operators.

Triangularize the factor and normalize the primal graph

Following Yuan and Wu, introduce the invertible signal transformation

$$L_{\Psi} := \begin{bmatrix} I & 0 \\ \Psi_{21} & \Psi_{22} \end{bmatrix}, \quad \begin{bmatrix} z \\ \tilde{w} \end{bmatrix} = L_{\Psi} \begin{bmatrix} z \\ w \end{bmatrix}.$$

Moving this transformation from the factor to the interconnection gives

$$\Psi_{\Delta} := \Psi L_{\Psi}^{-1} = \begin{bmatrix} \Psi_{11} - \Psi_{12}\Psi_{22}^{-1}\Psi_{21} & \Psi_{12}\Psi_{22}^{-1} \\ 0 & I \end{bmatrix}.$$

Hence the triangular form is obtained by an invertible change of internal signals; it is not an additional restriction on the original factor. Moreover,

$$\Psi \begin{bmatrix} z \\ w \end{bmatrix} = \Psi_{\Delta} \begin{bmatrix} z \\ \tilde{w} \end{bmatrix}.$$

On the graph $z = Pw$, the same transformation gives

$$L_{\Psi} \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} P \\ F_0 \end{bmatrix} w = \begin{bmatrix} PF_0^{-1} \\ I \end{bmatrix} F_0 w.$$

Consequently,

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w = \Psi_{\Delta} \begin{bmatrix} PF_0^{-1} \\ I \end{bmatrix} F_0 w = \begin{bmatrix} G_0 \\ I \end{bmatrix} F_0 w,$$

where

$$G_0 = (\Psi_{11}P + \Psi_{12})(\Psi_{21}P + \Psi_{22})^{-1}.$$

Since $\tilde{w} = F_0 w$, the normalized system is

$$\tilde{z} = G_0 \tilde{w}. \tag{21}$$

The primal IQC is now an ordinary gain condition:

$$\begin{aligned}\mathcal{I}_T(w) &:= \left\langle p_T \Psi \begin{bmatrix} P \\ I \end{bmatrix} w, J_{z,w} p_T \Psi \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle_{L_{\mathbf{H}}} \\ &= \|p_T G_0 F_0 w\|^2 - \|p_T F_0 w\|^2.\end{aligned}$$

If $\|G_0\| \leq \rho < 1$, the bounded causal invertibility of F_0 assumed at the start gives

$$\mathcal{I}_T(w) \leq -\frac{1-\rho^2}{\|F_0^{-1}\|^2} \|p_T w\|^2.$$

In a state-space representation, L_Ψ induces the corresponding invertible congruence transformation between the original filtered realization and the normalized realization.

Obtain the paired dual transformation

Define

$$\hat{J} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathbf{D}(\Psi) := (\hat{J}^* \Psi^\top \hat{J})^{-1}.$$

Calculate the dual filter directly. Since

$$\hat{J}^* \Psi^\top \hat{J} = \begin{bmatrix} \Psi_{22}^\top & -\Psi_{12}^\top \\ -\Psi_{21}^\top & \Psi_{11}^\top \end{bmatrix},$$

and

$$(\Psi_{\Delta,11})^\top = \Psi_{11}^\top - \Psi_{21}^\top (\Psi_{22}^\top)^{-1} \Psi_{12}^\top,$$

the block inverse formula gives

$$\mathbf{D}(\Psi) = \begin{bmatrix} (\Psi_{22}^\top)^{-1} + (\Psi_{22}^\top)^{-1} \Psi_{12}^\top ((\Psi_{\Delta,11})^\top)^{-1} \Psi_{21}^\top (\Psi_{22}^\top)^{-1} & (\Psi_{22}^\top)^{-1} \Psi_{12}^\top ((\Psi_{\Delta,11})^\top)^{-1} \\ ((\Psi_{\Delta,11})^\top)^{-1} \Psi_{21}^\top (\Psi_{22}^\top)^{-1} & ((\Psi_{\Delta,11})^\top)^{-1} \end{bmatrix}.$$

The desired paired factorization can now be read off directly:

$$\underline{\Psi}_\Delta := \begin{bmatrix} (\Psi_{22}^\top)^{-1} & (\Psi_{22}^\top)^{-1} \Psi_{12}^\top \\ 0 & I \end{bmatrix}, \quad \underline{L}_\Psi := \begin{bmatrix} I & 0 \\ \mathbf{D}(\Psi)_{21} & \mathbf{D}(\Psi)_{22} \end{bmatrix}.$$